



LV.0 → LV.1 · AI ENGINEER

The 5 I's that move you from **Level 0** → **1** AI engineer.

Inference, Instruction Tuning, In-context Learning and more. **Plain-English** definitions, one swipe each, plus a nation's AI mission and the minds to follow.

INFERENCE • FOUNDATIONS

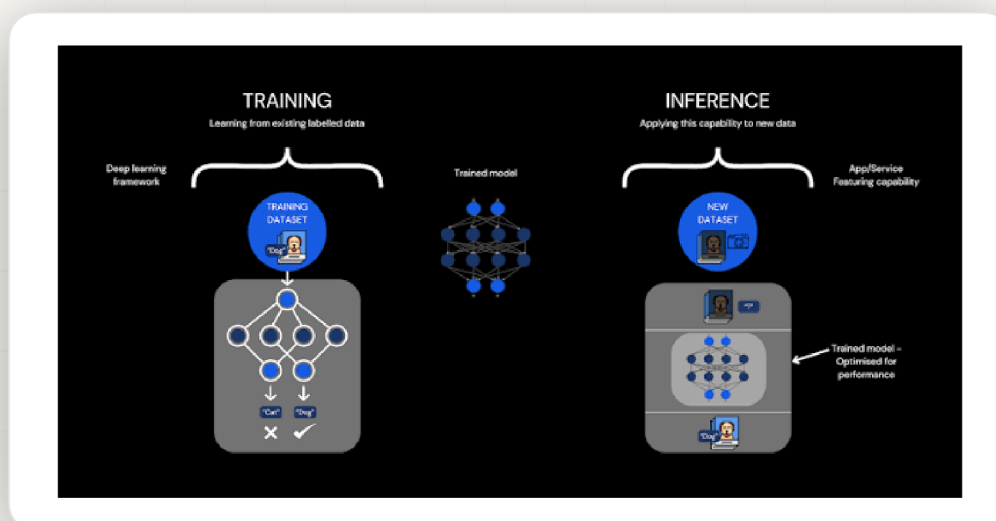
Inference

trained model • → • prediction

Training is school; inference is the job. Once a model is trained, inference is it **actually working**, classifying, predicting, or generating from new input. It is also where latency and cost suddenly bite.

USED IN

- Model serving
- Latency / cost
- Edge AI



TRAINING → INFERENCE

INSTRUCTION TUNING • LEARNING

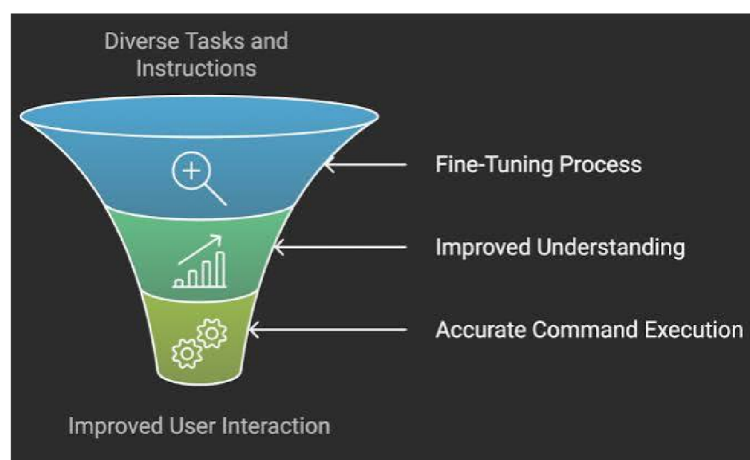
Instruction Tuning

predict text • → • follow commands

A raw language model just predicts text. Instruction tuning teaches it to **follow orders**, training on instructions paired with good responses until it can answer, summarize, and code on request.

USED IN

- Chat models
- RLHF pipelines
- Assistants



TASKS → COMMAND-FOLLOWING

IN-CONTEXT LEARNING • INFERENCE

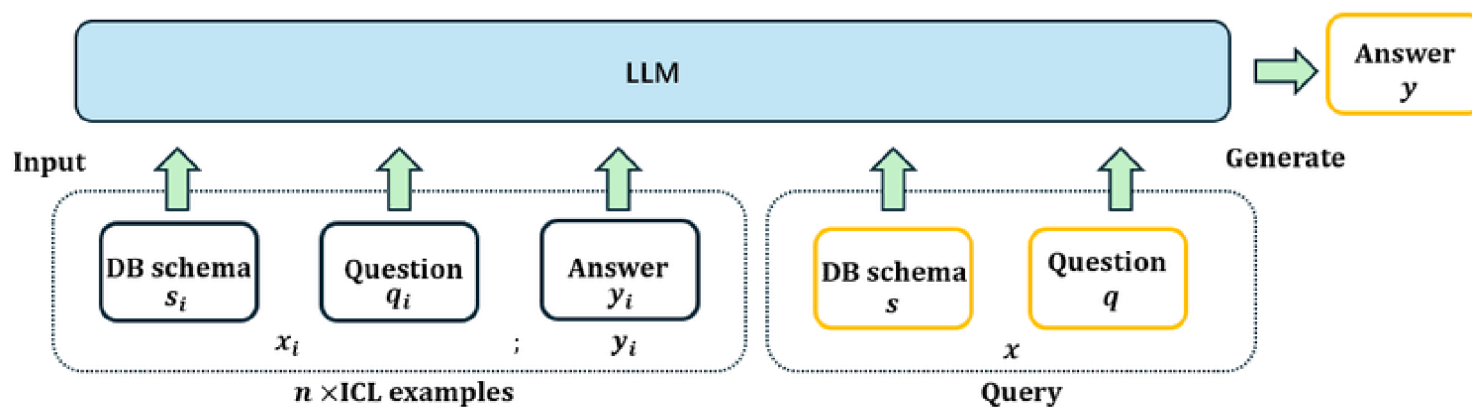
In-context Learning

learn • from • the prompt

No weights change, yet the model still learns. Drop a few examples in the prompt and it picks up the task on the spot. That trick, **learning from context alone**, is what makes few-shot prompting work.

USED IN

- Few-shot prompting
- Prompt design
- LLM apps



EXAMPLES IN-CONTEXT

Imputation

fill • the • gaps

Missing data, meet your replacement. Rather than tossing incomplete rows, imputation **estimates the gaps** with averages, neighbors, or a model, keeping your dataset whole so training stays clean.

USED IN

- Data cleaning
- Feature prep
- KNN / MICE

28	5.1
?	4.7
34	?

→

28	5.1
31	4.7
34	4.9

ESTIMATE MISSING CELLS

INDEXING • INFRASTRUCTURE

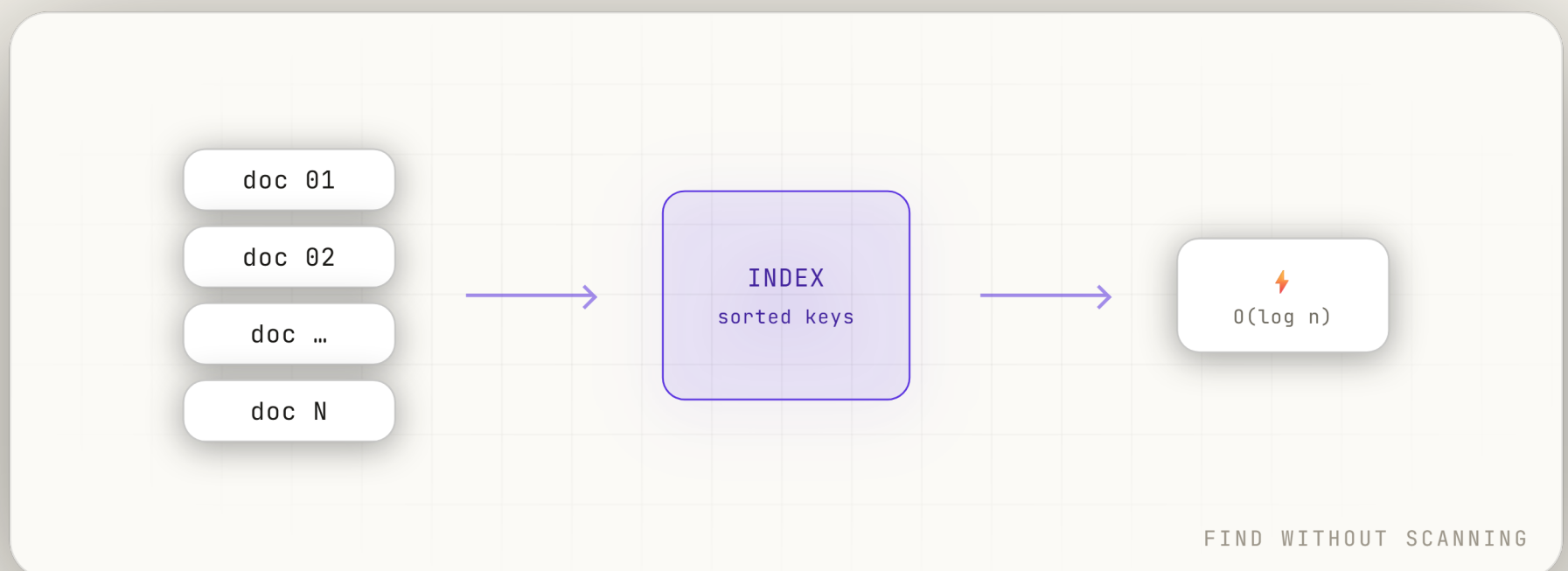
Indexing

organize • for • fast lookup

Finding one record by reading them all does not scale.
Indexing **organizes data so the right item surfaces fast**,
which is what makes search engines, databases, and vector
stores feel instant.

USED IN

- Search engines
- Databases
- Vector stores



↓ UP NEXT • INITIATIVE

Initiative next.

Beyond the **5 I-words**, the national mission building an entire country's AI stack from compute to talent.

NATIONAL AI MISSION

IndiaAI

compute • models • talent • governance



A country building its own AI stack.
IndiaAI is the national mission for **shared compute, homegrown models, and funding**, plus responsible-AI rules, aiming to put India firmly on the global map.

WHAT IT IS

India's national AI development mission.

FOCUS AREAS

GPUs, indigenous LLMs, skilling, safety.

WHY IT MATTERS

Compute, models & funding at nation scale.

PAIRS WITH

Startups, universities, public datasets.

TEXT-TO-IMAGE MODEL

Imagen

prompt • → • photorealistic image

Describe it and Imagen draws it.
Google's text-to-image family pairs strong language understanding with diffusion to turn **plain prompts into detailed, photoreal pictures.**

WHAT IT IS

Google's text-to-image diffusion model.

IN AI STACKS

Creative tools, marketing, ad pipelines.

WHY TEAMS USE IT

Photoreal images from natural prompts.

PAIRS WITH

Vertex AI, Gemini, the Google stack.

↓ UP NEXT • PEOPLE & A CHANNEL

Spotlight next.

Two **foundational minds** and a **high-signal channel** worth a follow to keep your I-stack current.

HIGH-SIGNAL, LOW-NOISE

Ilya Sutskever

co-founder • OpenAI & SSI



Few people shaped modern AI more. From AlexNet to seq2seq to OpenAI's GPT line, Ilya kept **being in the room where it happened**, and now builds safe superintelligence at SSI.

WHAT THEY POST

Deep learning, scaling, AI safety.

FIND THEM

X @ilyasut, talks & interviews.

WHY FOLLOW

One of the architects of modern AI.

BEST FOR

Big-picture thinking on where AI heads.

HIGH-SIGNAL, LOW-NOISE

Ian Goodfellow

inventor of GANs • deep learning



The man who taught networks to compete. Ian invented GANs, sparking the first wave of realistic AI imagery, and **literally wrote the textbook** on deep learning.

KNOWN FOR

GANs & the *Deep Learning* textbook.

WORKED AT

Google Brain, OpenAI, Apple, DeepMind.

WHY FOLLOW

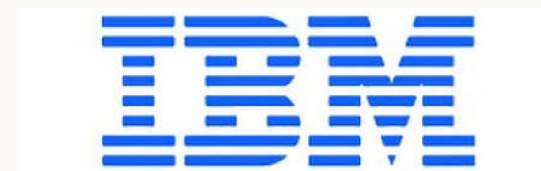
Pioneer of modern generative models.

BEST FOR

Foundations of generative AI.

HIGH-SIGNAL, LOW-NOISE

IBM Techno



YouTube • whiteboard explainers

Whiteboard-clear explainers on the hard stuff. IBM's channel breaks **transformers, RAG, embeddings, and agents** into short, visual lessons that beginners and pros both keep open in a tab.

WHAT THEY POST

Visual explainers on AI & cloud.

FIND THEM

YouTube @IBMTechology.

WHY FOLLOW

Complex topics, whiteboard-simple.

BEST FOR

Grasping RAG, agents & LLM basics.

The I stack, in one card.

Save it for later. Drop a comment if there's an I-term you'd add.

01 **Inference** foundations

02 **Instruction Tuning** learning

03 **In-context Learning** inference

04 **Imputation** data

05 **Indexing** infrastructure

+ **IndiaAI** initiative

+ **Imagen** tool

@ **Ilya Sutskever** people

@ **Ian Goodfellow** people

@ **IBM Technology** channel



FOLLOW FOR THE FULL A TO Z

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J Next